

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures

GPA: 3.7/4.0

Projects

Office Hours Queue | *React/TypeScript, Go/Gin, PostgreSQL, Redis, Docker*

Sep 2025 – Present

Co-Founder · Advisor: Prof. Louis Oliphant (CS)

- Reduced student wait times from 1–3 hours to under 15 minutes by launching a full-stack queue platform serving 500+ students across 2 courses with real-time SSE updates, Redis pub/sub, and role-based portals.
- Migrating the backend from Python/Flask to Go/Gin, targeting 10x concurrent connections for real-time SSE streaming and Redis pub/sub via goroutines and channels.
- Decreased average answer lookup time to ~2s with an ML-powered knowledge assistant (FastAPI, ChromaDB, scikit-learn) using semantic search across 4 sources and automated FAQ generation.
- Enabled one-command deployments for 6 services by containerizing with Docker Compose and Traefik reverse proxy, automating CI/CD via GitHub Actions and Portainer webhooks.

E-Beam 3D Printer Control System | *Python, Node.js, Express, GCP*

Jan – Aug 2025

Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a multi-threaded Python control system coordinating 6+ subsystems with fault-tolerant backoff and graceful degradation, enabling 72+ hour unattended experiments.
- Engineered a real-time observability dashboard (Node.js, Express, GCP) enabling researchers to monitor multi-day experiments, with log2-based downsampling to visualize weeks of sensor data.
- Implemented a G9SP safety controller driver parsing 199-byte binary packets with bit-field decoding for 13 interlock points, checksum validation, and fail-safe recovery on sensor disconnects.

AI/ML Algorithm Implementations | *Python, PyTorch, scikit-learn*

Sep – Dec 2024

- Implemented 9 AI/ML algorithms from scratch including LeNet-5 CNN (100-class classification), Q-Learning/SARSA RL agents (30K episodes), PCA, A* search, and Naive Bayes.

Experience

Software Developer Intern (*Part-time*) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

- Built a neonatal screening lab system (React/TypeScript, FastAPI, PostgreSQL) processing 300+ samples daily with automated validation and abnormality flagging for 4 technicians.
- Accelerated clinical report validation by 85% with a batch pipeline (pdfplumber, regex) that cross-validates patient data from 100+ PDF reports across multiple formats.
- Automated 3+ hours of manual work into a single-click extraction pipeline (Python, pandas, xlswriter) transforming mass spectrometry data for 40+ compounds into formatted Excel reports.

Mainframe Intern (*Part-time*) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Reduced manual triage time by 90% with a Python-based usage tracker that parsed job logs into structured CSV, flagged remediation status, and prioritized follow-ups across 5+ teams.
- Deployed an IaC automation platform (Ansible, Terraform) adopted department-wide, streamlining 15+ batch job workflows and reducing manual operations across state systems.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison*

Sep 2024 – Aug 2025

- Mentored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval time by 70% with an Oracle ERP analytics tool (Python, oracledb, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.

Skills

Languages: TypeScript, JavaScript, Python, Go, Java, C/C++, SQL, Bash, HTML/CSS

Frameworks & Tools: React, Gin, Flask, FastAPI, Node.js/Express, Docker, Docker Compose, Terraform, Git, GitHub Actions

Data & ML: PyTorch, scikit-learn, ChromaDB, OpenAI API, pandas, NumPy, Matplotlib, Seaborn

Databases & Cloud: PostgreSQL, Redis, Oracle DB, MySQL, SQLite, GCP, Linux

Other: REST APIs, SSE, WebSockets, Monitoring/Observability, OAuth2, JWT, CI/CD, Agile/Scrum, Pytest